

SAM MEYDANSHAH

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (Honours)

2021 – 2026

- Coursework: Algorithms, Operating Systems, Object-Oriented Programming, Functional Programming, Database Management, App Development, User Interfaces, Data Structures, Compilers, Shell Scripting

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, Rust, JavaScript, TypeScript, Kotlin, Swift, SQL

Frameworks & Tools: PyTorch, TensorFlow, LangChain, React, Angular, Next.js, Node.js, Django, FastAPI, Spring Boot, Gradle, Docker, Kubernetes, AWS, GCP, Supabase, PostgreSQL, MongoDB, Firebase, JWT, LLMs, CI/CD

EXPERIENCE

Tenstorrent

January 2026 – Present

Software Engineer Intern

Toronto, Ontario, CA

- Optimized the tensor tilize operator by refactoring data transformation calls, **making the operation by 77% faster**
- Diagnosed & fixed L1 data corruption across **9 dataflow kernels** caused by incorrect circular-buffer pointer usage
- Optimized the reshape operator by eliminating intermediate sharded-to-interleaved and interleaved-to-sharded conversions for tiled inputs, **reducing operation latency by 21%**
- Authored 2D fabric tests covering CCL operations and integrated them into Blackhole post-commit and nightly pipelines, ensuring continuous correctness coverage for distributed multi-chip LLM inference
- Built a memory sanitizer for AI accelerator kernels, catching silent hardware bugs (OOB writes, buffer errors, synchronization violations, etc) in emulation, catching errors **across over 4,000 tests, covering 14 unique kernels**

Payments Canada

May 2025 – August 2025

Software Developer Intern

Ottawa, Ontario, CA

- Developed and optimized participant search functionality (Angular + Spring Boot) in the RTR payments system, improving query flexibility and reducing errors for **100+ financial institutions**
- Redesigned API responses across RTR and FIF systems to return richer payloads, reducing frontend API calls and **improving load performance by 25%**
- Implemented robust data validation and error-handling in RTR and FIF APIs, **resolving 4 recurring 500-level errors** and preventing duplicate submissions or intentionally manipulated user inputs

Teranet

January 2024 – April 2024

Software Developer Intern

Toronto, Ontario, CA

- Implemented core components of GeoWarehouse, **optimizing load times for 100,000+** real estate professionals
- Optimized existing search query logic **improving query efficiency by 25%**

Teranet

January 2023 – April 2023

Software Engineer Intern

Toronto, Ontario, CA

- Built a full-stack internal dashboard using Angular, TypeScript, and Node.js to **monitor 30+ backend services**, improving DevOps visibility and **cutting manual status checks by 75%**

TECHNICAL PROJECTS

🔗 **DocChat** | React, FastAPI, pgvector, OpenAI API, LangChain, pdfplumber, Docker

- Full-stack Q&A app implementing a complete RAG pipeline including PDF ingestion, text chunking, and semantic retrieval to answer natural language queries grounded in uploaded documents

🔗 **AI Real Estate Assistant** | Node.js, Express, Claude API, Vanilla JS, Render

- Built and deployed an AI assistant in production for a real estate client, integrating the Claude API

🔗 **YouTube Wrapped** | Kotlin, Exposed, Jetpack Compose, Supabase, Google Cloud

- Developed an Android app in Kotlin to generate Spotify-Wrapped-style stats from users' YouTube watch history

🔗 **Biquadris** | C/C++, XWindows

- Engineered a "Tetris" variant from scratch with custom rendering using object oriented programming principles